Data Science

Data Visualization for Exploration

GENERAL ASSEMBLY

Exploratory Data Analysis

Lesson roadmap

- Welcome + Warm-up
- Visualizing relationships
- Visualizing trends
- Using `matplotlib`'s object-oriented interface
- Advanced visualizations with `seaborn`

Learning objectives

1. Apply appropriate chart types when exploring a data set.
2. Explore patterns and relationships in data with bar charts and scatter plots.
3. Create line charts to explore temporal trends in a data set.
4. Understand the object-oriented way of working with `matplotlib` to fully control your visualizations.
5. Generate heat maps in `seaborn`.

Visualizing relationships

What are the relationships within our data?

Relationships could mean:

- Differences among groups
  - Sales by region
- Comparing metrics
  - Median salary and median bonus amount

→ **Bar charts**

- Looking for correlations between our variables
  - How does the size of a house relate to the asking price?
- Investigating outliers
  - Which houses are too expensive relative to their size?

→ **Scatter plots**

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Let's see it for ourselves!

Let's open **02-visualizing-relationships.ipynb**

Visualizing trends

What are the trends in our data?

Trends suggest **time series** data.

- Daily stock price over time
- Weekly sales data

To visualize time series data, we need:

- Data that is **represented as dates** (Not strings that look like dates!)
- Data at the right **level of granularity** (Daily? Weekly? Monthly?)
- Line charts

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Let's see it for ourselves!

Let's open **03-visualizing-trends.ipynb**

Break time

Using matplotlib's object-oriented interface

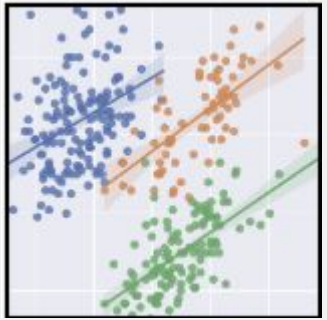
Let's see it for ourselves!

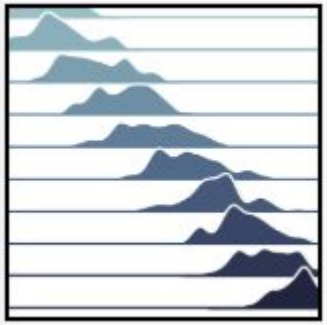
Let's open **04-advanced-matplotlib.ipynb**

Advanced visualizations with seaborn

Using another visualization library

Sometimes matplotlib doesn't have the visualization we're looking for. One alternative is seaborn.





Images from https://seaborn.pydata.org/examples/index.html

Let's see it for ourselves!

Let's open **04-advanced-matplotlib.ipynb**

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Data Visualization for Exploration

Summary

Key takeaways

Visualizing relationships and trends

- Bar charts and scatter plots let us explore relationships within our data
- Line charts help visualize temporal trends

Choosing appropriate charts

- Bar charts for categorical comparisons
- Scatter plots for correlations
- Line charts for trends over time
- Heatmaps for multi-dimensional relationships

Using matplotlib

- `matplotlib` has different “interfaces”
- The object-oriented interface is the preferred method
- Creating charts and figures directly gives us full control over our visualizations

Advanced chart types

- `seaborn` contains many advanced chart types
- The `seaborn` library is closely integrated with pandas and matplotlib

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Alternative visualization libraries

The Python ecosystem has even more visualization libraries! Notable ones include:

- plotly (for interactive visualizations)
- Altair ("declarative" visualizations, a different way of working with plots)
- plotnine (more familiar for users of R and ggplot)

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